

The nonseparable C^* -Krein–Milman theorem

Finite BW localization and pure extensions

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Status. `candidate_full_solution_likely_valid`. The argument answers one explicit open question in [2]; it should receive expert review before being treated as established literature.

The source question

For a unital C^* -algebra $\mathcal{A}$ and a separable Hilbert space $\mathcal{H}$, write

$$S_{\mathcal{H}}(\mathcal{A}) = \text{UCP}(\mathcal{A}, \mathcal{B}(\mathcal{H})).$$

The bounded weak (BW) topology is pointwise weak-operator convergence. Bhat and Kumar prove that $S_{\mathcal{H}}(\mathcal{A})$ is the BW-closure of the C^* -convex hull of its C^* -extreme points when $\mathcal{A}$ is separable or a type I factor. On source-PDF page 19 they state that the result is unknown for general nonseparable C^* -algebras [2, Section 5].

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So a generalized Krein–Milman theorem for $S_{\mathcal{H}}(\mathcal{A})$ would be to ask whether $S_{\mathcal{H}}(\mathcal{A})$ is the closure of the C^* -convex hull of its C^* -extreme points in BW-topology. Here the C^* -convex hull of any subset K of $S_{\mathcal{H}}(\mathcal{A})$ is given by

$$\left\{ \sum_{i=1}^n T_i^* \phi_i(\cdot) T_i; \phi_i \in K, T_i \in \mathcal{B}(\mathcal{H}) \text{ with } \sum_{i=1}^n T_i^* T_i = I_{\mathcal{H}} \right\}. \quad (5.1)$$

The goal of this section is to prove a Krein–Milman type theorem for $S_{\mathcal{H}}(\mathcal{A})$, whenever $\mathcal{A}$ is a separable C^* -algebra or $\mathcal{A}$ is of the form $\mathcal{B}(\mathcal{G})$ for some Hilbert space $\mathcal{G}$. The proof of these two cases are different. Note that $\mathcal{B}(\mathcal{G})$ is not separable, when $\mathcal{G}$ is infinite dimensional. As yet we do not know the result in full generality (i.e. for non-separable C^* -algebras). We recall here that as mentioned before such a theorem can be found in [12] for $S_{\mathcal{H}}(\mathcal{A})$, when $\mathcal{H}$ is a finite dimensional Hilbert space and $\mathcal{A}$ is an arbitrary C^* -algebra, and in [4] for general $\mathcal{H}$ and commutative C^* -algebra $\mathcal{A}$. Thus our result provides an important development towards this theorem in infinite dimensional Hilbert space settings.

Lemma 5.1. *Let $\phi \in S_{\mathcal{H}}(\mathcal{A})$ be such that $\phi(a) = \sum_{n \geq 1} \phi_n(a)$ in WOT, for all $a \in \mathcal{A}$, where $\{\phi_n : \mathcal{A} \rightarrow \mathcal{B}(\mathcal{H})\}_{n \geq 1}$ is a countable family of pure completely positive maps. Then ϕ is in the BW-closure of C^* -convex hull of C^* -extreme points of $S_{\mathcal{H}}(\mathcal{A})$.*

Proof. We assume that the collection $\{\phi_n\}_{n \geq 1}$ in the sum of ϕ is countably infinite. The finite case follows similarly and easily. For each $n \geq 1$, let $(\pi_n, V_n, \mathcal{K}_n)$ be the minimal Stinespring triple for ϕ_n . Then each π_n is irreducible, as ϕ_n is pure by hypothesis. Note that

$$\sum_n V_n^* V_n = \sum_n \phi_n(1) = \phi(1) = I_{\mathcal{H}}, \quad \text{in WOT.}$$

Proof intuition

BW approximation is local: a basic neighborhood tests only finitely many elements of $\mathcal{A}$, and those elements generate a separable subalgebra $\mathcal{B}$. The source theorem therefore supplies a C^* -convex approximation on $\mathcal{B}$.

The subtle point is extension. An arbitrary C^* -extreme map on $\mathcal{B}$ need not be known to have a C^* -extreme extension. Fortunately, the source proof uses much more structured extreme maps. Each is built from a pure UCP map and a one-dimensional pure compression of that same map. Pure UCP maps have pure extensions. Extending the first map and taking the same vector compression recreates the structured block on $\mathcal{A}$, where the source's direct-sum theorem again makes it C^* -extreme. Thus the same operator coefficients lift the local approximation to all of $\mathcal{A}$.

Two extension lemmas

Recall that a completely positive map α is *pure* if $0 \leq \gamma \leq \alpha$ implies $\gamma = t\alpha$ for some $t \in [0, 1]$.

Lemma 1 (pure extension). *Let $\mathcal{B} \subseteq \mathcal{A}$ be unital C^* -algebras and let $\alpha : \mathcal{B} \rightarrow \mathcal{B}(\mathcal{K})$ be pure and unital completely positive. Then α has a pure UCP extension $\tilde{\alpha} : \mathcal{A} \rightarrow \mathcal{B}(\mathcal{K})$.*

Proof. By Arveson's extension theorem [1], the set

$$\mathcal{F} = \{\beta \in \text{UCP}(\mathcal{A}, \mathcal{B}(\mathcal{K})) : \beta|_{\mathcal{B}} = \alpha\}$$

is nonempty. It is BW-compact and convex. It is also a face of $\text{UCP}(\mathcal{A}, \mathcal{B}(\mathcal{K}))$: if $t\beta_1 + (1-t)\beta_2 \in \mathcal{F}$, with $0 < t < 1$, then $t\beta_1|_{\mathcal{B}} \leq \alpha$. Purity gives $t\beta_1|_{\mathcal{B}} = s\alpha$; evaluation at 1 gives $s = t$, hence $\beta_1|_{\mathcal{B}} = \alpha$, and similarly for β_2 .

Choose an extreme point $\tilde{\alpha}$ of $\mathcal{F}$. Since $\mathcal{F}$ is a face, it is linearly extreme in the full UCP set. Suppose $0 \leq \gamma \leq \tilde{\alpha}$. Purity of the restriction gives $\gamma|_{\mathcal{B}} = s\alpha$. Thus $\gamma(1) = sI$. If $0 < s < 1$, then

$$\tilde{\alpha} = s(s^{-1}\gamma) + (1-s)((1-s)^{-1}(\tilde{\alpha} - \gamma))$$

is a convex decomposition into UCP maps. Linear extremality forces $\gamma = s\tilde{\alpha}$. The cases $s = 0, 1$ follow because a CP map vanishing at the unit is zero. Hence $\tilde{\alpha}$ is pure. $\square$

The next lemma isolates precisely the blocks appearing in the proof of [2, Lemma 5.1].

Lemma 2 (lifting the source blocks). *Let $\mathcal{K} \subseteq \mathcal{H}$, let $\alpha : \mathcal{B} \rightarrow \mathcal{B}(\mathcal{K})$ be pure UCP, and let $h \in \mathcal{K}$ be a unit vector. Put*

$$\vartheta(b) = \langle h, \alpha(b)h \rangle, \quad \xi(b) = \alpha(b) \oplus \vartheta(b)I_{\mathcal{K}^\perp}.$$

Assume (as for the source blocks) that ϑ is a pure state. Then ξ has a C^ -extreme UCP extension $\tilde{\xi} : \mathcal{A} \rightarrow \mathcal{B}(\mathcal{H})$.*

Proof. Take a pure extension $\tilde{\alpha}$ from Lemma 1 and define $\tilde{\vartheta}(a) = \langle h, \tilde{\alpha}(a)h \rangle$. The minimal Stinespring representation of a pure CP map is irreducible [1]. Every nonzero vector in an irreducible representation is cyclic; therefore $\tilde{\vartheta}$ is a pure state and extends ϑ .

Set

$$\tilde{\xi}(a) = \tilde{\alpha}(a) \oplus \tilde{\vartheta}(a)I_{\mathcal{K}^\perp}.$$

The scalar summand is a compression of the same irreducible Stinespring representation as $\tilde{\alpha}$. Their Stinespring ranges are nested, and the one-type multiplicity nest has countable completion. The direct-sum criterion [2, Theorem 3.6] therefore makes ξ C^* -extreme. Plainly $\xi|_{\mathcal{B}} = \xi$. $\square$

The same lemma includes the fixed residual block needed below: choose a pure state ω of $\mathcal{B}$. Then $b \mapsto \omega(b)I_{\mathcal{H}}$ is a one-type amplification, hence C^* -extreme by the same direct-sum criterion, and a pure extension of ω gives a C^* -extreme extension of this block.

Main theorem

Theorem 1. *Let $\mathcal{A}$ be any unital C^* -algebra and let $\mathcal{H}$ be separable. Then*

$$S_{\mathcal{H}}(\mathcal{A}) = \overline{\text{co}_{C^*}(\text{Cext } S_{\mathcal{H}}(\mathcal{A}))}^{\text{BW}}.$$

Thus the full-generality question in [2] has an affirmative answer.

Proof. Fix $\phi \in S_{\mathcal{H}}(\mathcal{A})$ and a basic BW neighborhood $\mathcal{N}$ of ϕ . It is enough to consider finitely many tests

$$|\langle (\psi(a_r) - \phi(a_r))u_s, v_s \rangle| < \varepsilon \quad (1 \leq r \leq m, 1 \leq s \leq q).$$

Let $\mathcal{B} = C^*(1, a_1, \dots, a_m)$. This is a separable unital C^* -subalgebra, so the separable case of [2, Theorem 5.3] applies to $\phi_0 = \phi|_{\mathcal{B}}$.

We use one feature visible in the proof of that theorem. Its Case I first approximates ϕ_0 by a compression of a countable direct sum of irreducible representations. That compression is a WOT-sum of pure CP maps. The proof of [2, Lemma 5.1] then approximates it by finite C^* -convex combinations

$$\eta = \sum_{j=0}^N T_j^* \xi_j(\cdot) T_j, \quad \sum_{j=0}^N T_j^* T_j = I_{\mathcal{H}}, \quad (1)$$

where every non-residual ξ_j is, up to a unitary, exactly a block from Lemma 2. The residual coefficient tends strongly to zero and its map may be chosen as $\omega(\cdot)I_{\mathcal{H}}$ for a pure state ω of $\mathcal{B}$. Choosing the two approximation stages sufficiently accurately gives η satisfying all the tests defining $\mathcal{N}$.

By Lemma 2, extend each ξ_j to a C^* -extreme $\tilde{\xi}_j \in S_{\mathcal{H}}(\mathcal{A})$; use the pure-state amplification extension for the residual block. Keep the coefficients in (1) and define

$$\tilde{\eta} = \sum_{j=0}^N T_j^* \tilde{\xi}_j(\cdot) T_j.$$

This belongs to the C^* -convex hull of the C^* -extreme points of $S_{\mathcal{H}}(\mathcal{A})$, and $\tilde{\eta}|_{\mathcal{B}} = \eta$. Since every tested a_r lies in $\mathcal{B}$, the map $\tilde{\eta}$ belongs to $\mathcal{N}$.

Every basic BW neighborhood of every ϕ therefore meets the stated hull, which is precisely the asserted BW-closure identity. $\square$

Verification, limitations, and novelty check

The proof deliberately does not assert that arbitrary C^* -extreme maps extend. It lifts only the pure/compression blocks actually constructed in the source’s Lemma 5.1. The compact extension-fiber argument uses nets and needs no metrizability. The conclusion is local BW closure, so no diagonal sequence over all of a nonseparable algebra is required.

The result retains the source convention that $\mathcal{H}$ is separable. It removes the separability and type I assumptions on $\mathcal{A}$; it does not address the paper’s other questions about the structure of normal C^* -extreme maps or whether every C^* -extreme map is linearly extreme.

A bounded novelty search on 13 August 2026 covered the run indexes, the parsed arXiv corpus, exact-title citations, and primary-source searches for the full nonseparable BW theorem and the pure-extension route. It found later work on specialized classes but no statement of Theorem 1. This supports candidate novelty but is not an exhaustive priority claim.

References

- [1] W. Arveson, *Subalgebras of C^* -algebras*, Acta Math. 123 (1969), 141–224. In particular, the CP extension and pure-extension machinery.
- [2] B. V. Rajarama Bhat and M. Kumar, *C^* -extreme maps and nests*, arXiv:2103.09600; J. Funct. Anal. 282 (2022), Paper No. 109397.